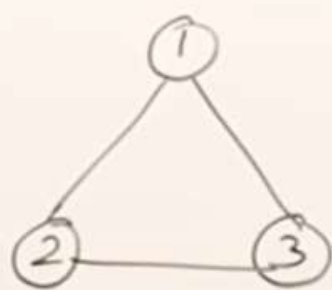
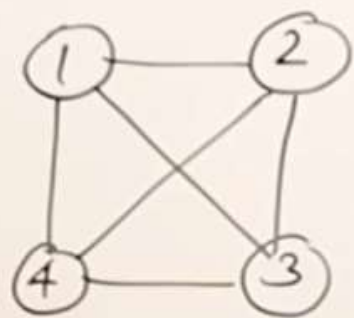


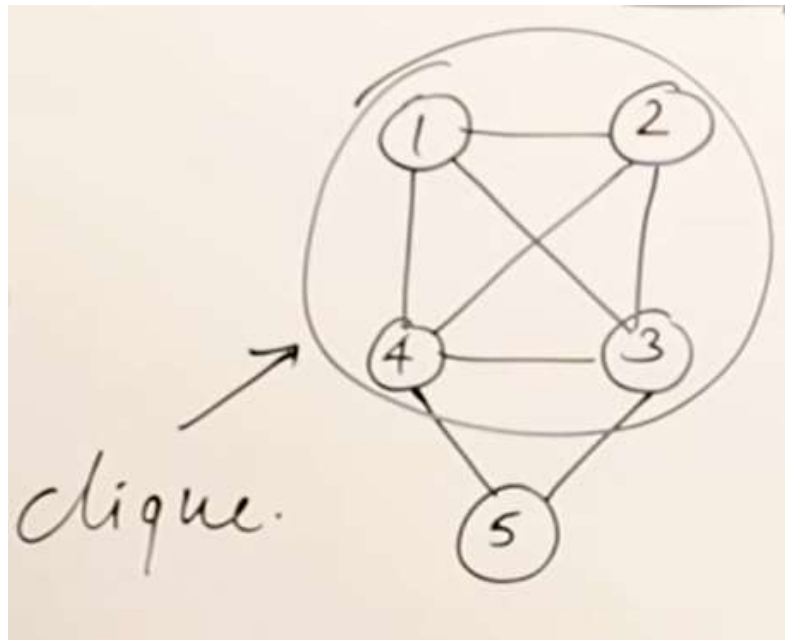
NP-Hard Graph Problem - Clique Decision Problem (Reducibility)

<https://www.youtube.com/watch?v=qZs767KQcvE>

Clique Decision Problem (CDP)



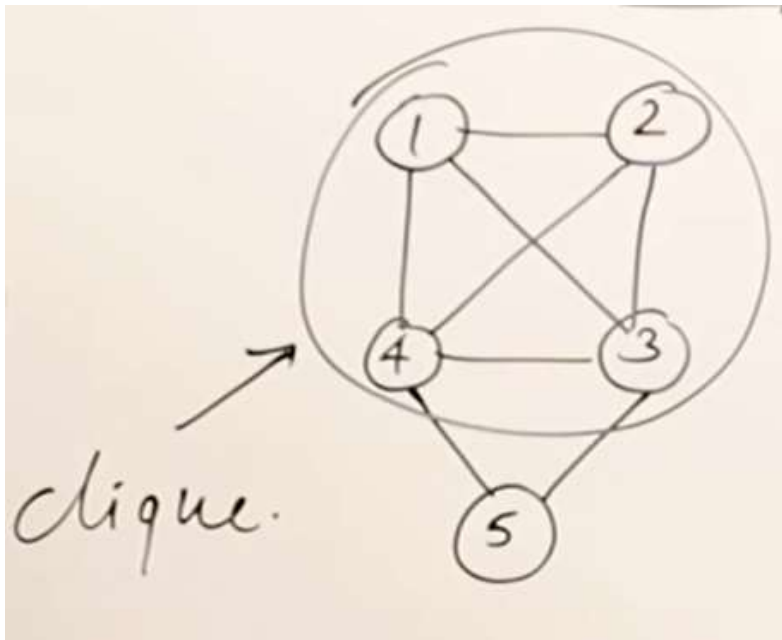
1. what is clique?
2. what are Decision & Optimization Problems
3. what are NP-Hard Graph Problems
4. Procedure to Prove NP-Hard
5. Prove CDP is NP-Hard.



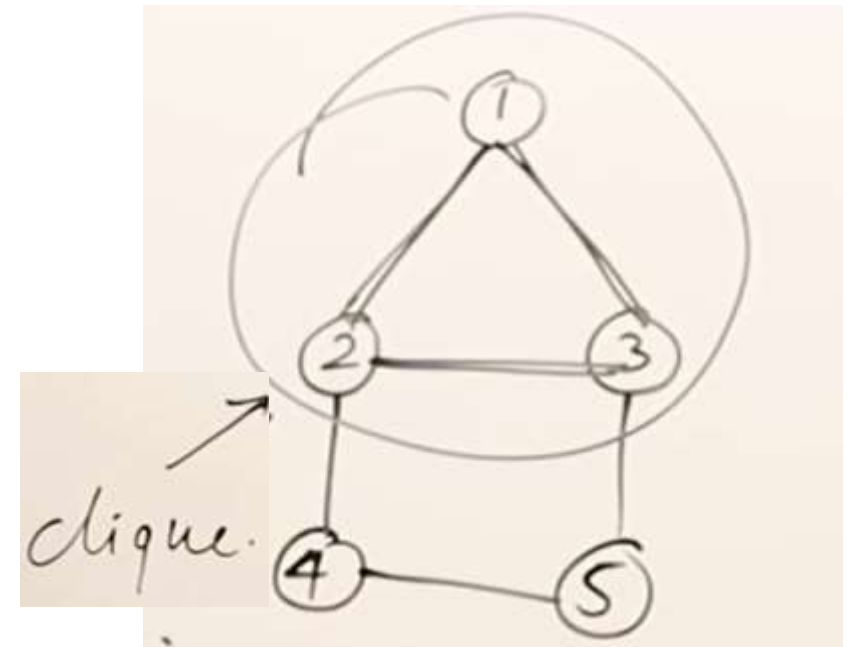
A **subgraph** of a graph which is actually a complete graph is called **clique**

Complete Graph

$$|V| = n$$
$$|E| = \frac{n(n-1)}{2}$$



A **subgraph** of a graph which is actually a complete graph is called **clique**



Is it having a **complete graph** with 4 vertices?

Yes
Is it having a **clique** of size 4 in graph?

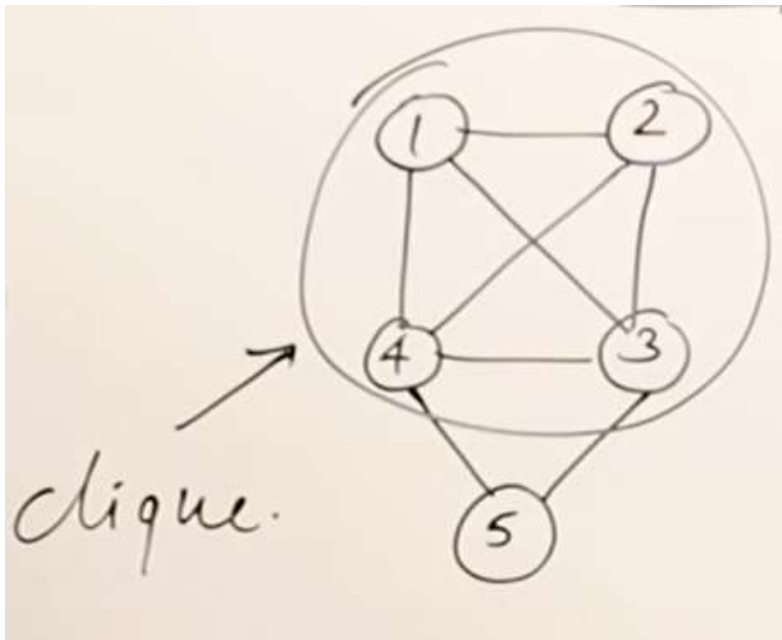
Complete Graph

$$|V| = n$$

$$|E| = \frac{n(n-1)}{2}$$

There are 5 vertices in a graph.
Is it having a **complete graph** with 4 vertices?

NO,
Since 2→5 and 3→4 are not connected



A **subgraph** of a graph which is actually a complete graph is called **clique**

Is it having a complete graph with 4 vertices?

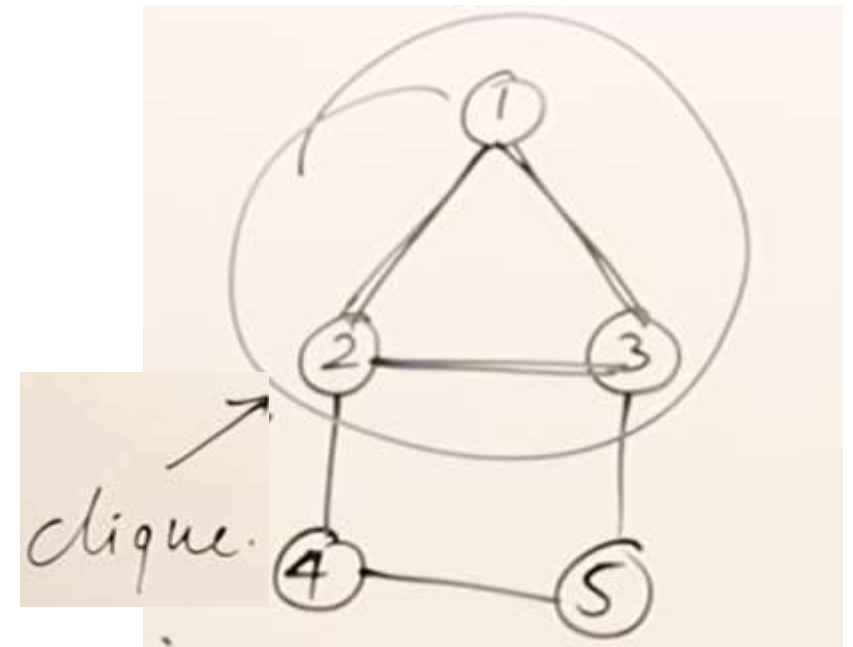
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Is it having a clique of size 4 in graph?

Complete Graph

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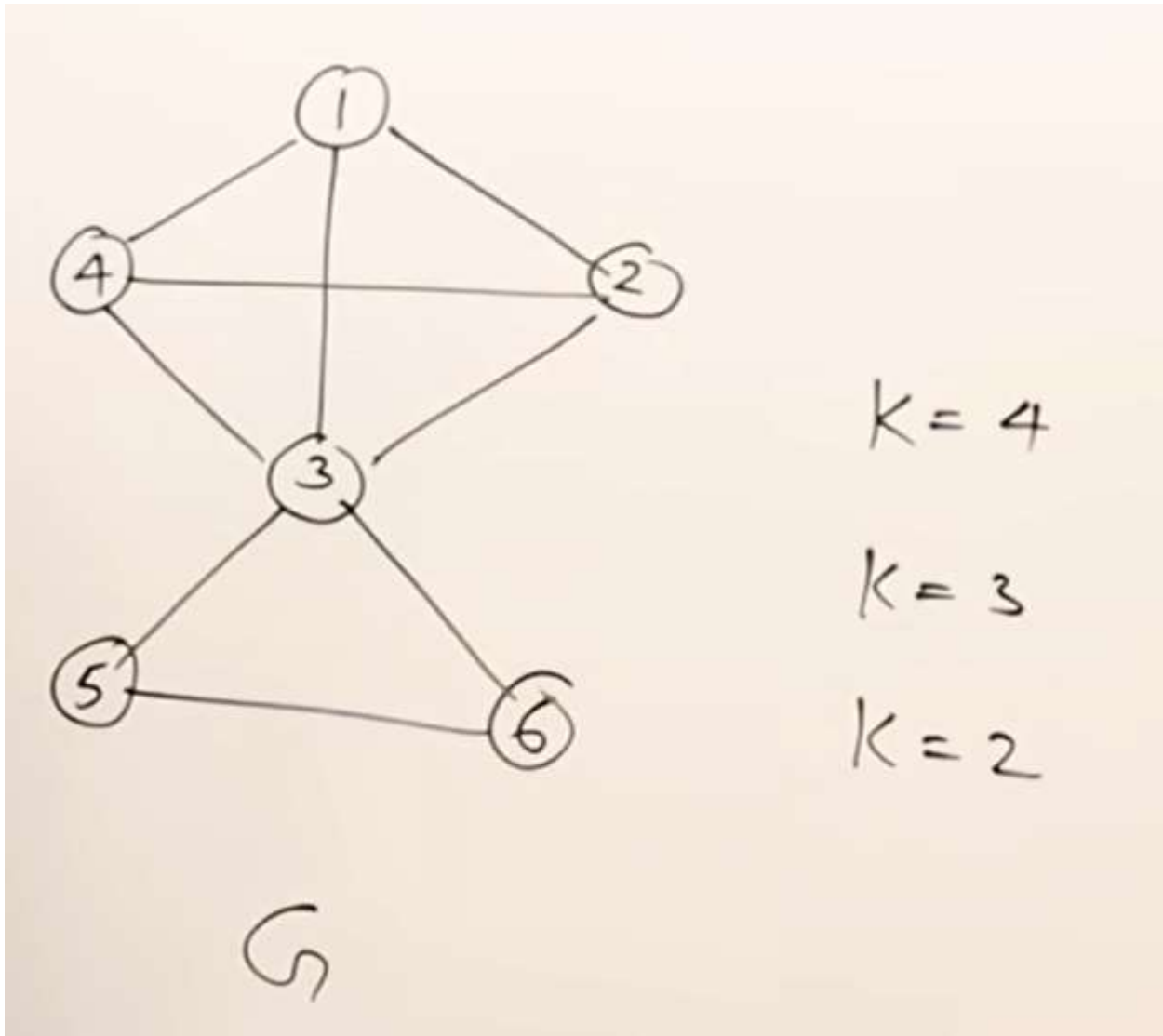
$$|E| = \frac{n(n-1)}{2}$$



There are 5 vertices in a graph.

Is it having a complete graph with **3** vertices?

Yes

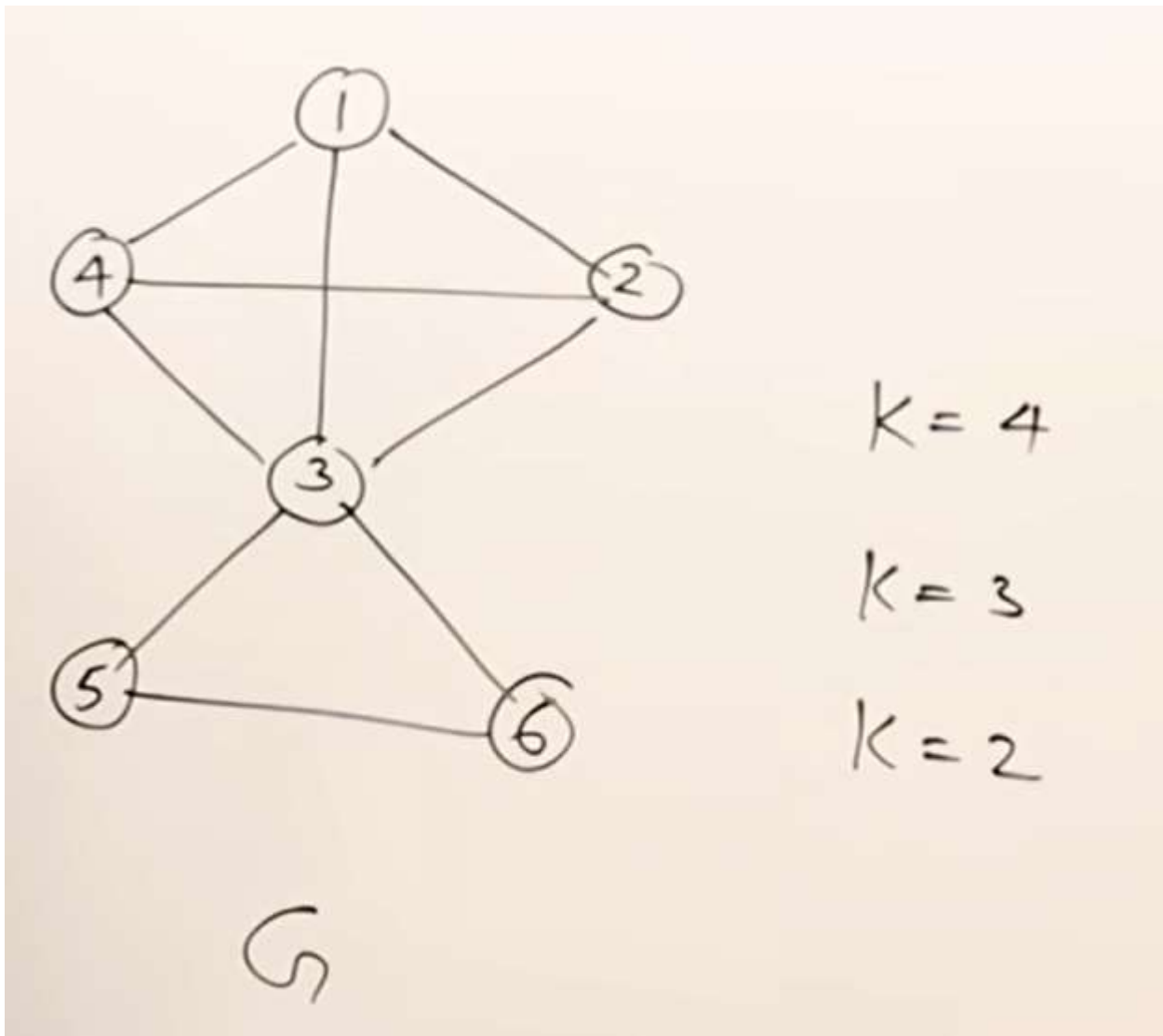


Is there any sub graph which is forming a complete graph?

Yes

There are 4 vertices $\langle 1, 2, 3, 4 \rangle$ which forms a complete graph, $k = 4$

- Finding a maximum clique is **Optimization Problem**
- Finding a clique of size k is present in the graph or not **is Decision problem**



Is there any sub graph which is forming a complete graph?

Yes

(i) Vertices (1,2,3,4) -
clique of size 4 forms a complete graph

(ii) Vertices (3,5,6) -
clique of size 3 forms a complete graph

Vertices (1,2,3) -
clique of size 3
forms a complete graph

(iii) Vertices (5,6) -
clique of size 2

forms a complete graph

1. Finding a maximum clique : **Optimization Problem**. Eg. What is the maximum clique size in the graph?
2. Finding a clique of size k is present or not in the graph : **Decision problem**
: Yes or No

Clique Decision Problem

- It is graph problem
- **Decision problem** : it is to find out whether it is **true** or **false**
 - **The query: Is there any clique of size k in the graph?**
- Find if a graph is having a clique of size k
- Also, we have to prove that it is NP-hard
- If Decision problem is NP-hard, then Optimization is also NP-hard
- Both are NP-hard problems
- But we have to prove that "Clique Decision Graph Problem" is NP-hard

Procedure for proving Graph problem to be NP-hard

- Let us see What is a general method for proving a problem as NP-hard
- Let a problem be L_1 have to prove that the problem L_2 is NP-hard. $P = L_2$
- In order to prove that we have to select any problem L_1 which is already known as NP-hard and we have to show that L_1 reduces to L_2 . $L_1 \propto L_2$
- How will you prove it? We will take the example (Instance I_1) of L_1 problem and from example of L_1 , we will prepare an example (Instance I_2) of L_2 . $L_1 \propto L_2$
 $I_1 \quad I_2$

Procedure for proving Graph problem to be NP-hard

- If I_2 can be solved in Polynomial time then I_1 can also be solved in Polynomial time (or in other words if I_2 can be solved then I_1 can also be solved)
- The conversion from I_1 to I_2 is done in Polynomial time
- Example: What problem we want to show as NP-hard?

$$I_1 \quad \alpha \quad I_2$$

Procedure for proving Graph problem to be NP-hard

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- **Example:** What problem we want to show as NP-hard?
CDP
- What is the known problem that we have which is already NP-hard?

A handwritten diagram on a light brown background. It shows a sequence of three elements: I_1 , α , and I_2 . Below I_1 and I_2 are horizontal dashed lines. The symbol α is positioned between I_1 and I_2 , indicating a transformation or reduction from I_1 to I_2 .

Procedure for proving Graph problem to be NP-hard

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Procedure for proving Graph problem to be NP-hard

- If I_2 can be solved in Polynomial time then $L_1 \propto L_2$
also be solved in Polynomial time (or in other words I_1 can be solved then I_2 can also be solved)

- The conversion from I_1 to I_2 is done in Polynomial time

- **Example:** What problem we want to reduce to CDP

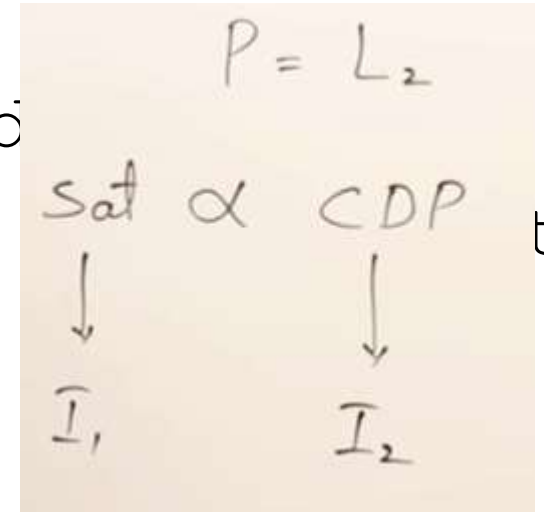
$$\begin{array}{ccc} \text{sat} & \propto & \text{CDP} \\ L_1 & \propto & L_2 \\ I_1 & & I_2 \end{array}$$

NP-hard

$$\begin{array}{ccc} P = L_2 & & \\ \text{sat} & \propto & \text{CDP} \\ \downarrow & & \downarrow \\ I_1 & & I_2 \end{array}$$

- What is the known problem that we have which already NP-hard? **SAT** (satisfiability)
- We have **to prove** that $\text{SAT} \propto \text{CDP}$
- For that, we have to take an example (**CNF**) of SAT and

- Let us prove that $\text{SAT} \propto \text{CDP}$
 - And hence prove that CDP is also NP-hard
 - SAT is the known NP-hard and now we are
- CDP is also NP-hard



Proof

- Let us prove that SAT $\propto$ CDP
- Take the formula with 3 variables for SAT
- Formula is made of 3 clauses and all k (here $k=3$) clauses are Anded

$$\begin{aligned} & \text{Sat} \propto \text{CDP} \\ & x_1, x_2, x_3 \\ & F = \bigwedge_{i=1}^K C_i \\ & F = \underbrace{(x_1 \vee x_2)}_{C_1} \wedge \underbrace{(\bar{x}_1 \vee \bar{x}_2)}_{C_2} \wedge \underbrace{(x_1 \vee x_3)}_{C_3} \end{aligned}$$

- From this formula, a **graph is prepared** such that it **should have a clique** and **that clique should be of size k** . (As per the given example here, $k = 3$)

- Don't mix all the vertices of different clauses. Keep it separately.
- If $x_1 \in C_1$, represented as $\langle x_1, 1 \rangle$
...
- *How?*

$\text{Sat} \propto \text{CDP}$
 x_1, x_2, x_3
 $F = (x_1 \vee x_2) \wedge (\bar{x}_1 \vee \bar{x}_2) \wedge (x_1 \vee x_3)$
 $\quad \quad C_1 \quad \quad \quad C_2 \quad \quad \quad C_3$
 $\quad \quad \langle \bar{x}_1, 2 \rangle \quad \langle \bar{x}_2, 2 \rangle$
 $\quad \quad \quad \circ \quad \quad \quad \circ$
 Clause C_1 $\langle x_1, 1 \rangle \circ$ Clause C_2 $\langle x_1, 3 \rangle \circ$
 $\langle x_2, 1 \rangle \circ$ Clause C_3 $\langle x_3, 3 \rangle \circ$

$$G \quad V = \{ \langle a, i \rangle \mid a \in c_i \}$$

- Don't mix all the vertices of different clauses. Keep it separately.
- What are the **vertices in the Graph**?
- In a graph G , a variable or literal $\langle a, i \rangle$ belongs to some clause C_i

$$\text{Sat} \propto \text{CDP}$$

$$x_1, x_2, x_3$$

$$F = \underbrace{(x_1 \vee x_2)}_{C_1} \wedge \underbrace{(\bar{x}_1 \vee \bar{x}_2)}_{C_2} \wedge \underbrace{(x_1 \vee x_3)}_{C_3}$$

$$\langle \bar{x}_1, 2 \rangle \quad \langle \bar{x}_2, 2 \rangle$$

$$\langle x_1, 1 \rangle \circ$$

$$\circ \langle x_1, 3 \rangle$$

$$\langle x_2, 1 \rangle \circ$$

$$\circ \langle x_3, 3 \rangle$$

- Let us see how to form edges?
- Rule 1: Should not connect the vertices of same clause
- Rule 2: Should not connect the clause with any vertex's Negation
- (Such edge is **not allowed**)

$$\text{Sat} \propto \text{CDP}$$

$$x_1, x_2, x_3$$

$$F = (x_1 \vee x_2) \wedge (\bar{x}_1 \vee \bar{x}_2) \wedge (x_1 \vee x_3)$$

$$\begin{matrix} C_1 & & C_2 & & C_3 \\ & \langle \bar{x}_1, 2 \rangle & \langle \bar{x}_2, 2 \rangle & & \\ & \bigcirc & \bigcirc & & \\ \langle x_1, 1 \rangle \bigcirc & & & \bigcirc \langle x_1, 3 \rangle & \\ \langle x_2, 1 \rangle \bigcirc & & & \bigcirc \langle x_3, 3 \rangle & \end{matrix}$$

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- Can I form an edge from $\langle x_1, 1 \rangle$ to $\langle x_2', 2 \rangle$

$$\text{Sat} \propto \text{CDP}$$

$$x_1, x_2, x_3$$

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$$x_1, x_2, x_3$$

$$F = (\underbrace{x_1}_{C_1} \vee \underbrace{x_2}_{C_2}) \wedge (\underbrace{\bar{x}_1}_{C_2} \vee \underbrace{\bar{x}_2}_{C_2}) \wedge (\underbrace{x_1}_{C_3} \vee \underbrace{x_3}_{C_3})$$

$$\begin{array}{ccc} \langle \bar{x}_1, 2 \rangle \circ & \langle \bar{x}_2, 2 \rangle \circ & \\ \langle x_1, 1 \rangle \circ & & \circ \langle x_1, 3 \rangle \\ \langle x_2, 1 \rangle \circ & & \circ \langle x_3, 3 \rangle \end{array}$$

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- $\langle x_2, 1 \rangle$ to $\langle x_2', 2 \rangle$?

$$\text{Sat} \propto \text{CDP}$$

$$x_1, x_2, x_3$$

$$F = (\underbrace{x_1 \vee x_2}_{C_1}) \wedge (\underbrace{\bar{x}_1 \vee \bar{x}_2}_{C_2}) \wedge (\underbrace{x_1 \vee x_3}_{C_3})$$

$\langle \bar{x}_1, 2 \rangle$ $\langle \bar{x}_2, 2 \rangle$
 $\circ$ $\circ$

$\langle x_1, 1 \rangle \circ$ $\circ \langle x_1, 3 \rangle$
 $\langle x_2, 1 \rangle \circ$ $\circ \langle x_3, 3 \rangle$

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- Can I form an edge from $\langle x_2, 1 \rangle$ to $\langle x_1', 2 \rangle$
- $\langle x_2, 1 \rangle$ to $\langle x_2', 2 \rangle$? **No,**
Not allowed

$$\text{Sat} \propto \text{CDP}$$

$$x_1, x_2, x_3$$

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$\langle x_1, 1 \rangle$ $\langle x_2, 1 \rangle$ $\langle x_1, 2 \rangle$ $\langle x_2, 2 \rangle$ $\langle x_1, 3 \rangle$ $\langle x_3, 3 \rangle$

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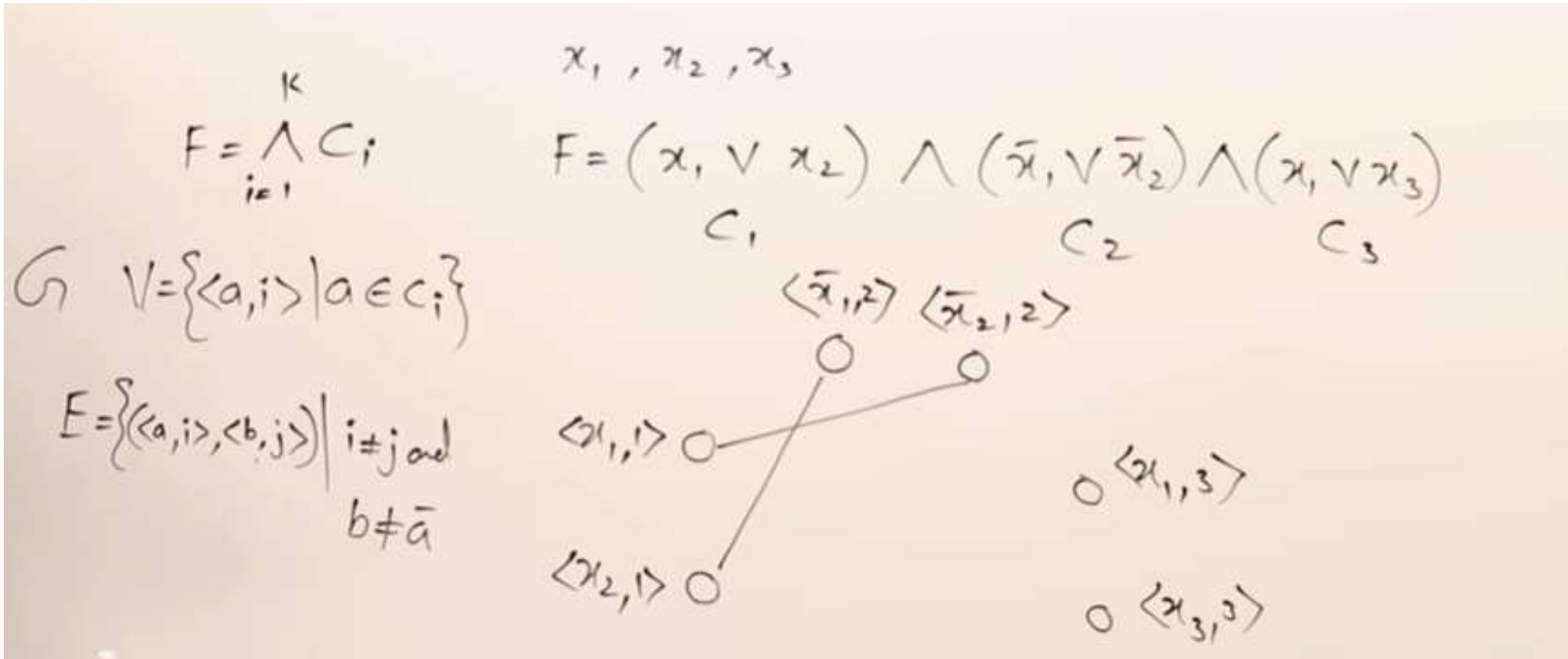
$$x_1, x_2, x_3$$

$$F = (\underbrace{x_1 \vee x_2}_{C_1}) \wedge (\underbrace{\bar{x}_1 \vee \bar{x}_2}_{C_2}) \wedge (\underbrace{x_1 \vee x_3}_{C_3})$$

$$\begin{array}{ccc} \langle \bar{x}_1, 2 \rangle \circ & \langle \bar{x}_2, 2 \rangle \circ & \\ \langle x_1, 1 \rangle \circ & & \circ \langle x_1, 3 \rangle \\ \langle x_2, 1 \rangle \circ & & \circ \langle x_3, 3 \rangle \end{array}$$

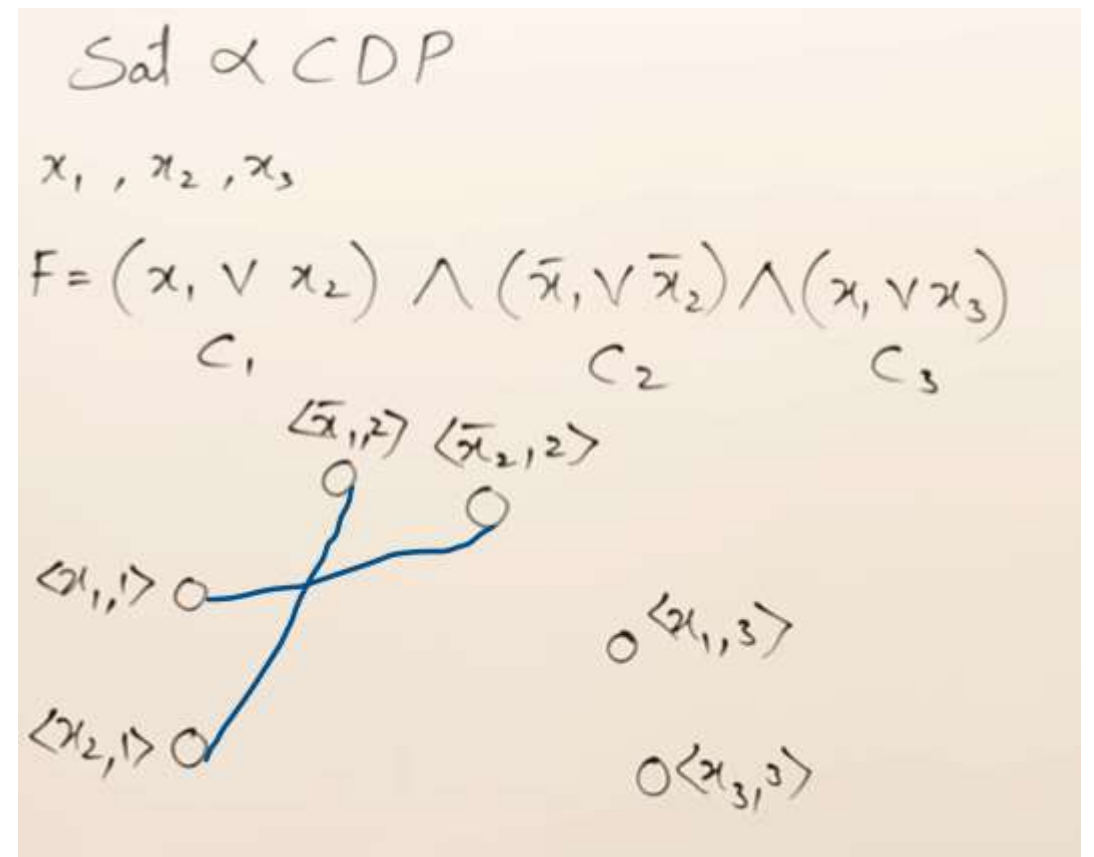
- Before completing the

- Before completing the graph, how will you form an edge?

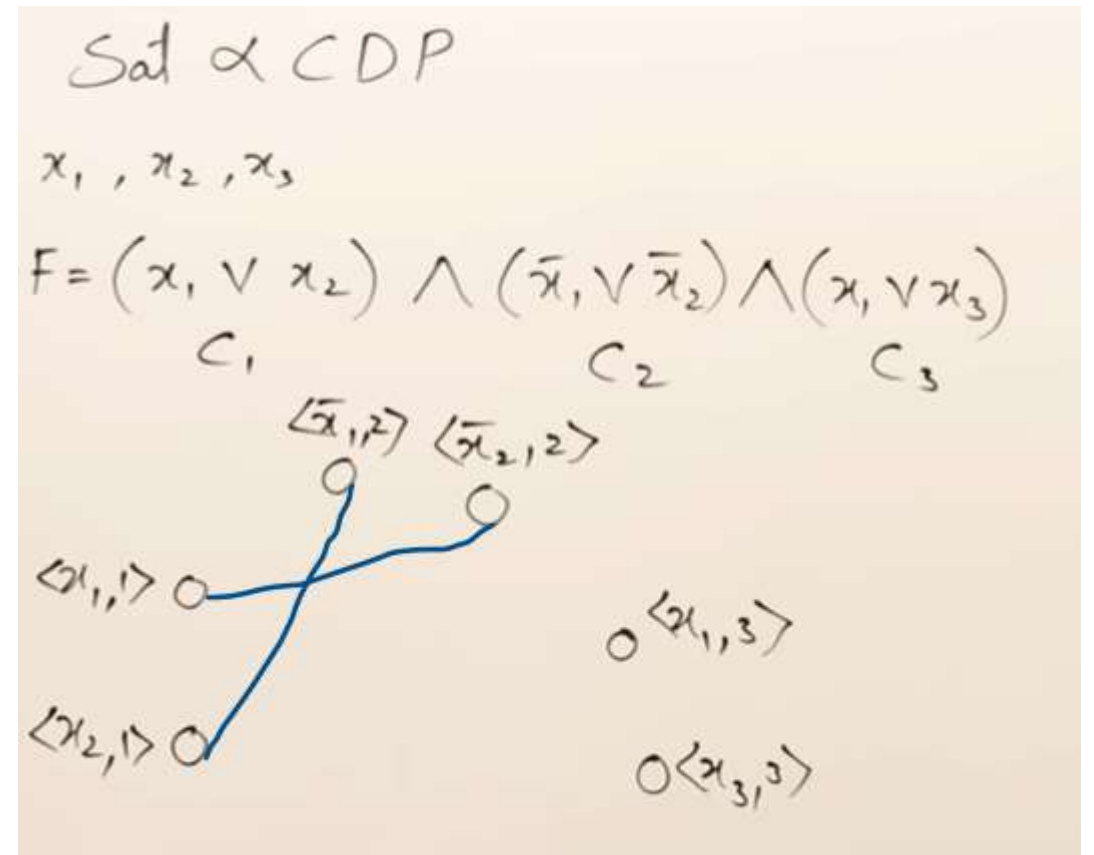


- These are the 2 conditions that are needed to take care while drawing edges

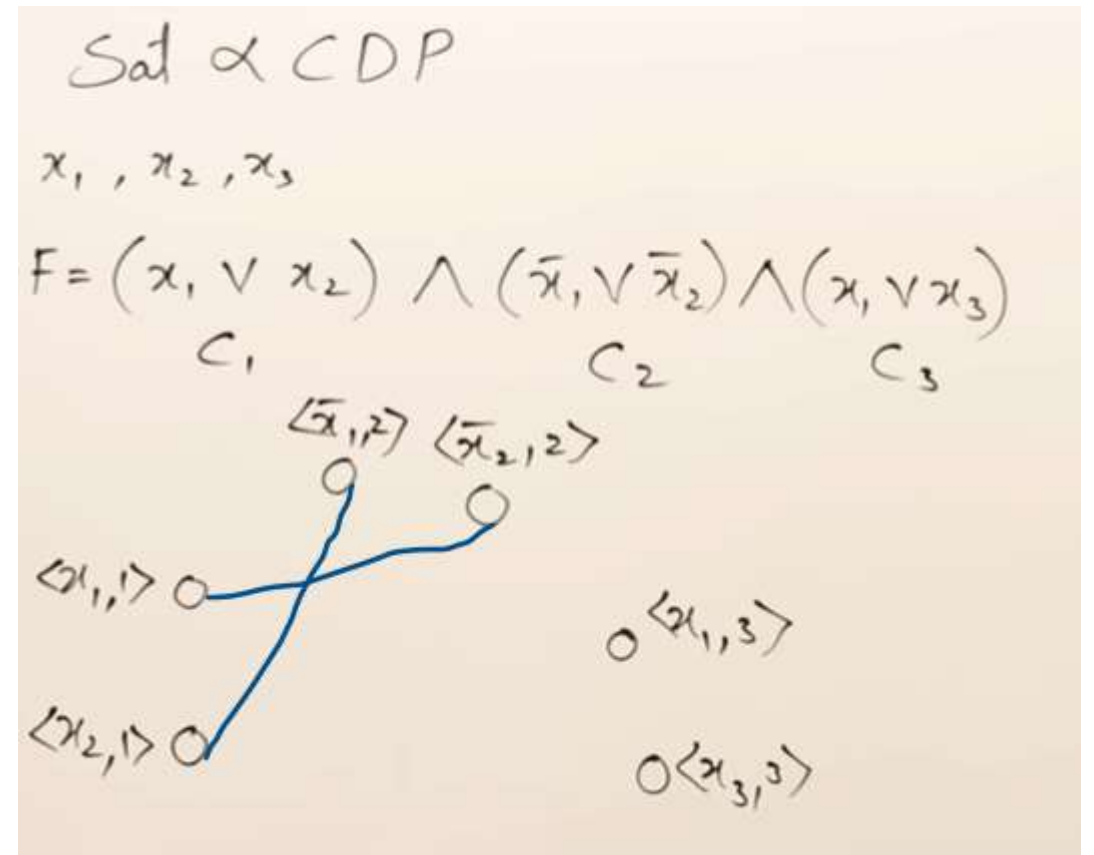
- Let us complete the graph, by drawing edges.
- Can I form an edge from $\langle x_1, 1 \rangle$ to $\langle x_1, 3 \rangle$ **yes**
- Can I form an edge from $\langle x_1, 1 \rangle$ to $\langle x_3, 2 \rangle$ **yes**



- Let us complete the graph, by drawing edges.
- Can I form an edge from $\langle x_1, 1 \rangle$ to $\langle x_1, 3 \rangle$ **yes**
- Can I form an edge from $\langle x_1, 1 \rangle$ to $\langle x_3, 2 \rangle$ **yes**
- Can I form an edge from $\langle x_2, 1 \rangle$ to $\langle x_1, 3 \rangle$ **yes**
- Can I form an edge from $\langle x_2, 1 \rangle$ to $\langle x_3, 3 \rangle$ **yes**
- Continue rest



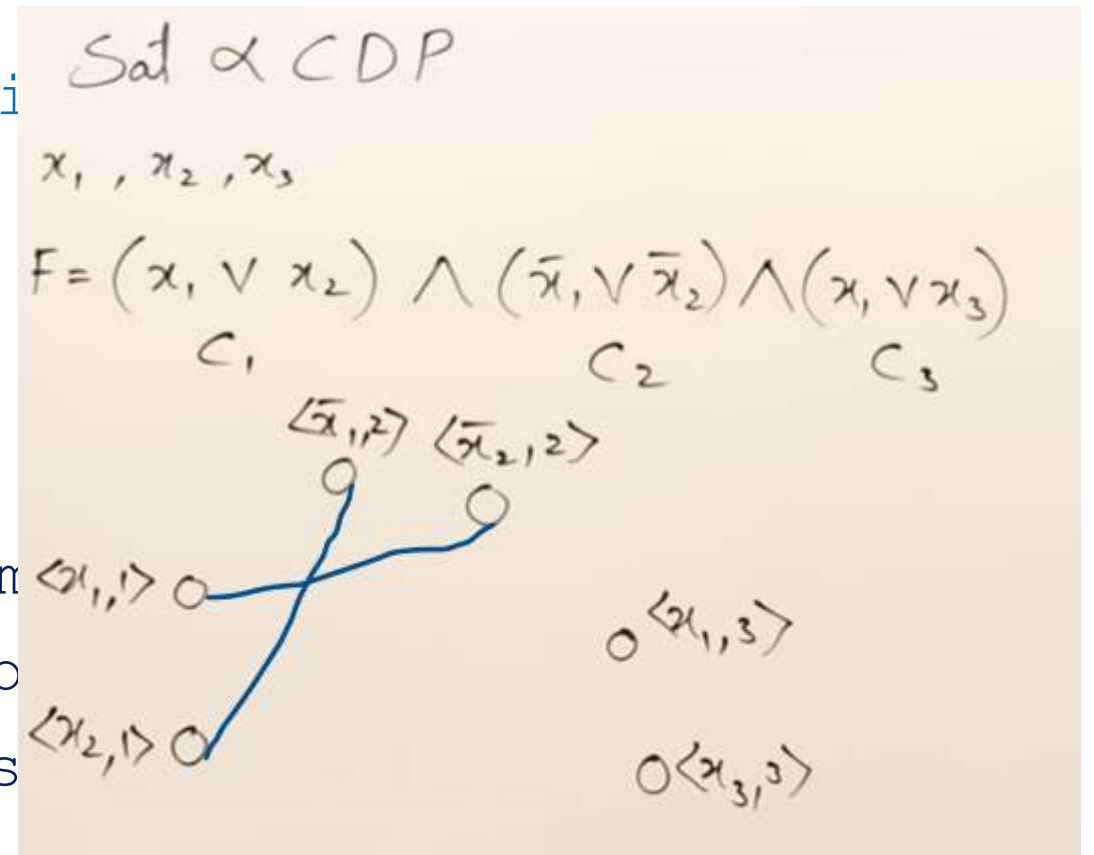
- Let us complete the graph, by drawing edges.
- Can I form an edge from $\langle x_1', 2 \rangle$ to $\langle x_1, 3 \rangle$ **No**
- Can I form an edge from $\langle x_1', 2 \rangle$ to $\langle x_3, 3 \rangle$ **yes**
- Can I form an edge from $\langle x_2', 2 \rangle$ to $\langle x_1, 3 \rangle$ **yes**
- Can I form an edge from $\langle x_2', 2 \rangle$ to $\langle x_3, 3 \rangle$ **yes**



- Let us complete the graph, by drawing
- Can I form an edge from $\langle x_1', 2 \rangle$ to
- Can I form an edge from $\langle x_1', 2 \rangle$ to
- Can I form an edge from $\langle x_2', 2 \rangle$ to
- Can I form an edge from $\langle x_2', 2 \rangle$ to
- So we got a graph which is an example of a graph for a formula F where F is SAT (satisfiable) and this graph is (applied) for this such that,

now if CDP is solved in polynomial time, then F can also be solved in polynomial time.

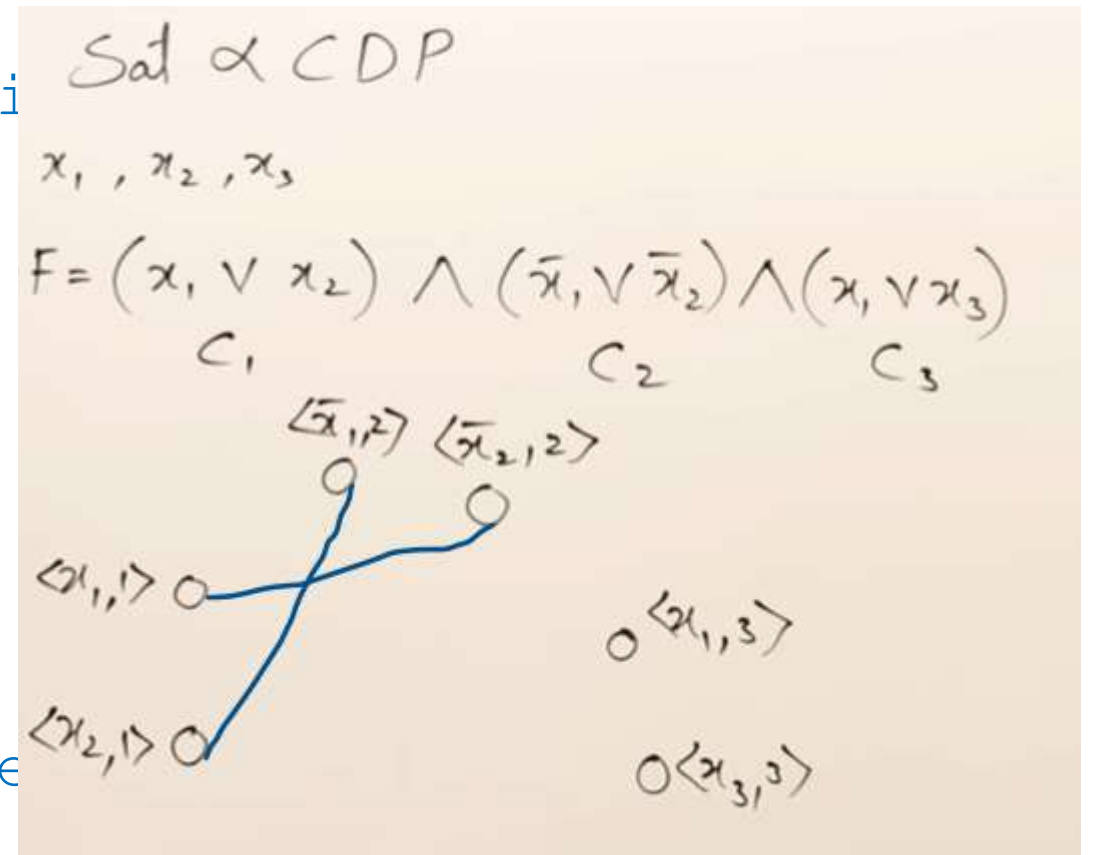
One more thing...



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One more thing...

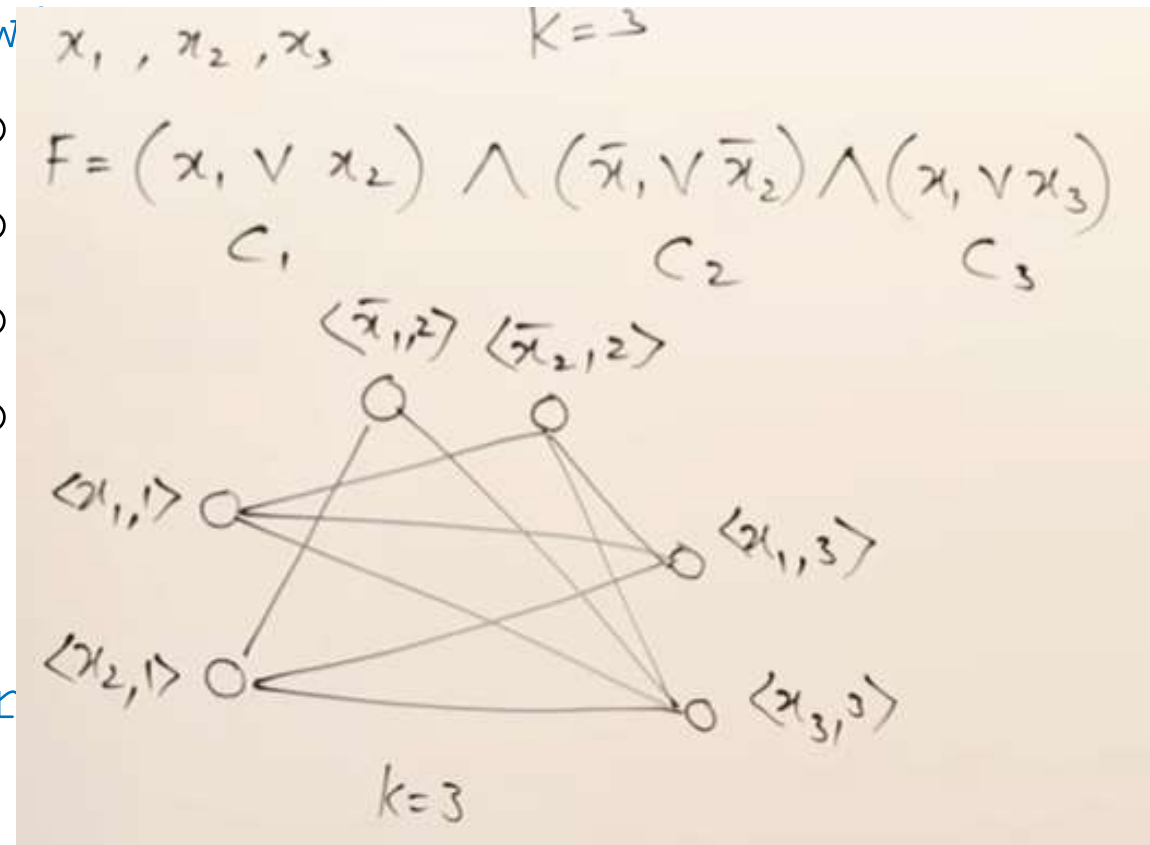
- What are the number of clauses here
- It is the size of k , $k = 3$
- Now find out, if we have a clique here?



- Let us complete the graph, by draw
- Can I form an edge from $\langle x_1', 2 \rangle$ to
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One more thing...

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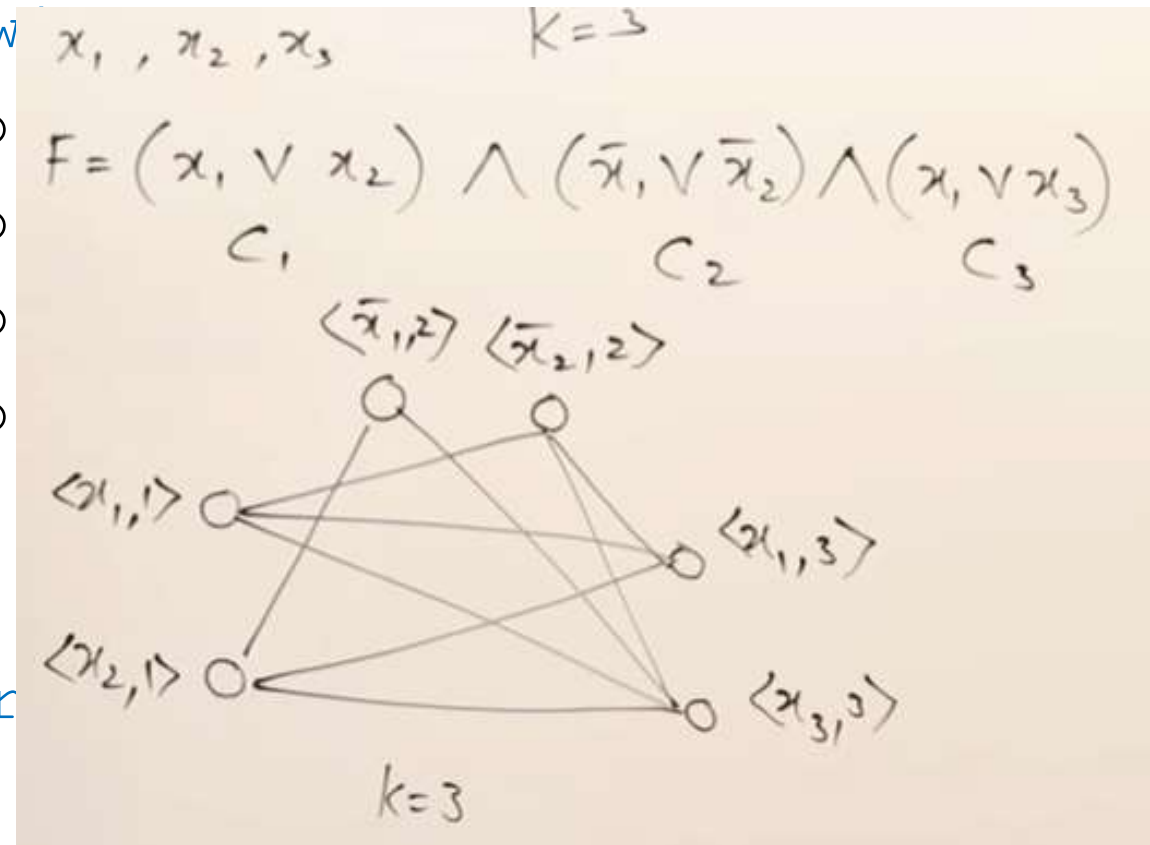


- Now find out, if we have a clique here?
- Yes. There is a clique of 2 [vertices $\langle x_2, 1 \rangle$ to $\langle x_1', 2 \rangle$]
- Also, there is a clique of 3 [vertices $\langle x_2, 1 \rangle$ to $\langle x_1', 2 \rangle$ to $\langle x_3, 3 \rangle$]
- Check do you have a clique of 4 here?

- Let us complete the graph, by draw
- Can I form an edge from $\langle x_1', 2 \rangle$ to
- Can I form an edge from $\langle x_1', 2 \rangle$ to
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One more thing...

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- Now find out, if we have a clique here?
- Yes. There is a clique of 2 [vertices $\langle x_2, 1 \rangle$ to $\langle x_1', 2 \rangle$]
- Also, there is a clique of 3 [vertices $\langle x_2, 1 \rangle$ to $\langle x_1', 2 \rangle$ to $\langle x_3, 3 \rangle$]
- Check do you have a clique of 4 here? **No**

- Let us complete the graph, by drawing edges.

- Can I form an edge from $\langle x_1', 2 \rangle$ to $\langle$

- Can I form an edge from $\langle x_1', 2 \rangle$ to $\langle$

- Can I form an edge from $\langle x_2', 2 \rangle$ to $\langle$

- Can I form an edge from $\langle x_2', 2 \rangle$ to $\langle$

There are more triangle which form cl

We can select any one of the triangle
shown in pink color $\Rightarrow$

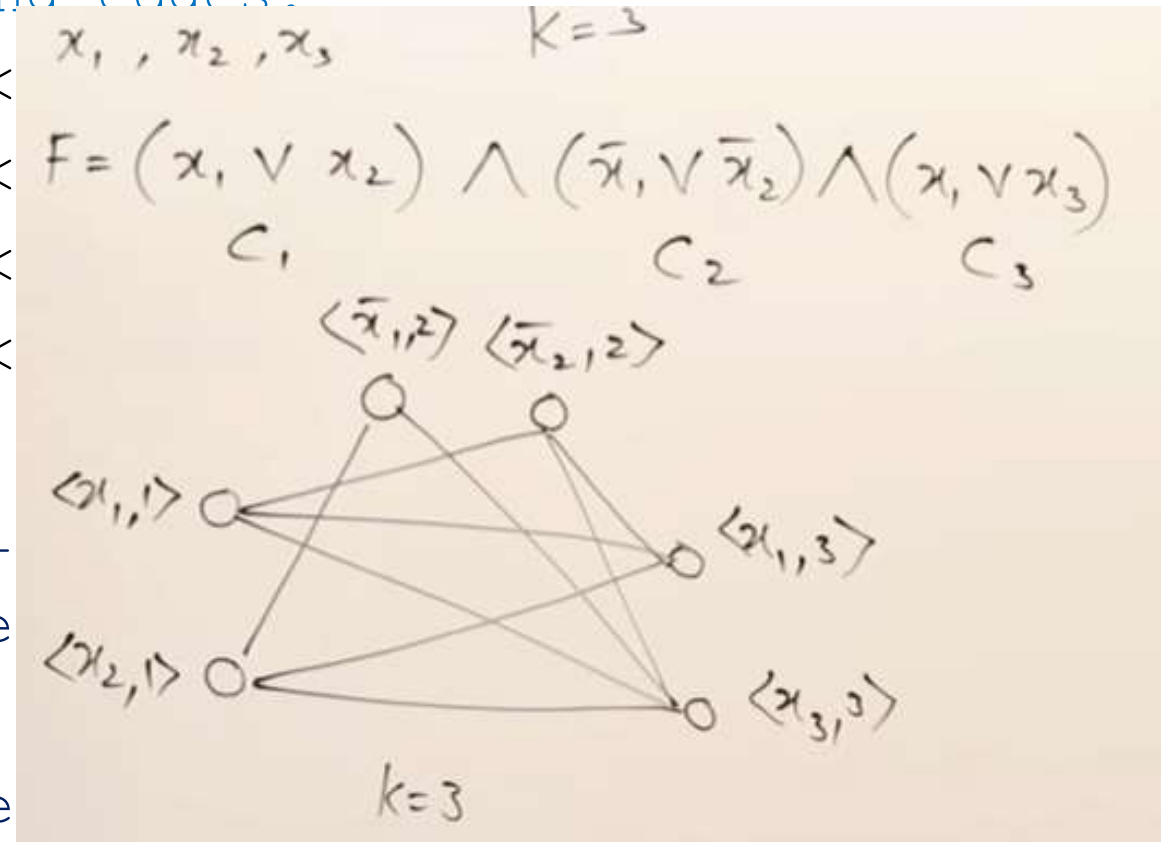
We have one clique of **size 3** bcoz the
is having **3 clauses**.

That's all.

Proof:

Proved here: If we take the formula **F** of **k clauses** then **k size clique**
will be there in the graph.

Reverse thing also can be solved. I.e. If **K size clique** is present in
graph, then Formula **F** can have **k clauses**.

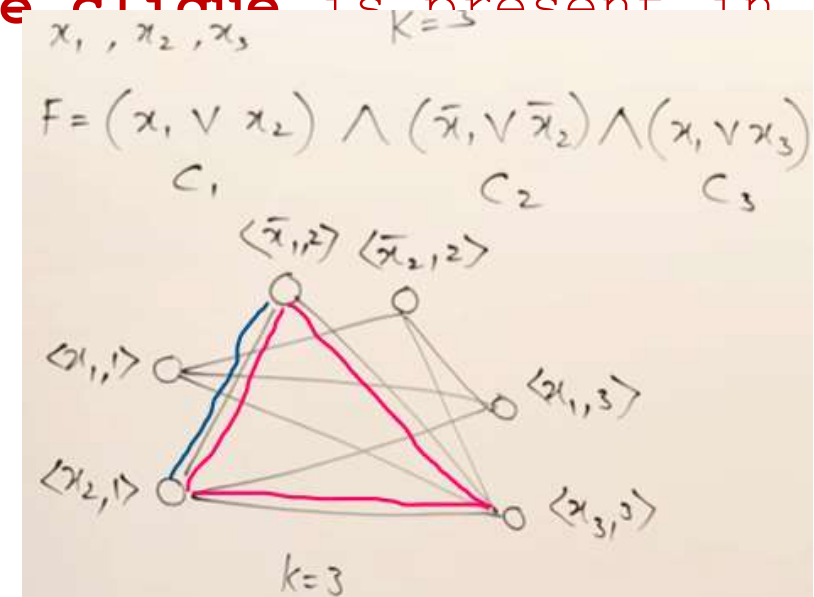


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- How many variable do we have? x_1, x_2, x_3
- Check the selected vertices (pink color).
- Mark them as TRUE (1)



x_1 x_2 x_3

x_1	x_2	x_3
0	1	1

x_1' is true. Hence
 x_1 is made 0

Proof:

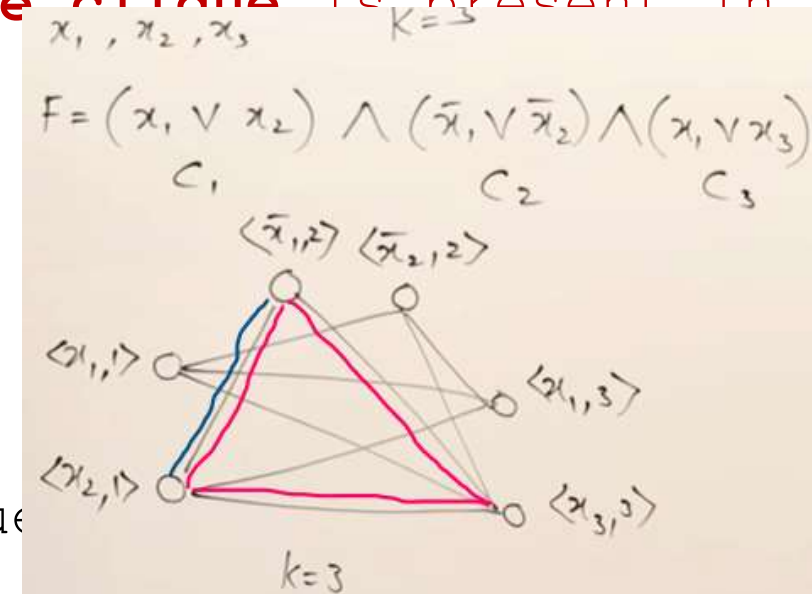
Proved here: If we take the formula **F** of **k clauses** then **k size clique** will be there in the graph.

Reverse thing also can be solved. I.e. If **K size clique** is present in graph, then Formula **F** can have **k clauses**.

- How many variable do we have? x_1, x_2, x_3
- Check the selected vertices (pink color).
- Mark them as TRUE (1)
- Substitute these values in Formula F, F is true
- Hence proved

$$F = (\overset{0}{x_1} \vee \overset{1}{x_2}) \wedge (\overset{1}{\bar{x}_1} \vee \overset{0}{\bar{x}_2}) \wedge (\overset{0}{x_1} \vee \overset{1}{x_3}) = 1$$

$C_1=1 \quad C_2=1 \quad C_3=1$



$$x_1 \quad x_2 \quad x_3$$

$$\begin{matrix} x_1 & x_2 & x_3 \\ 0 & 1 & 1 \end{matrix}$$

x_1' is true. Hence x_1 is made 0

- Try substituting the other values of other cliques here ?

Proof:

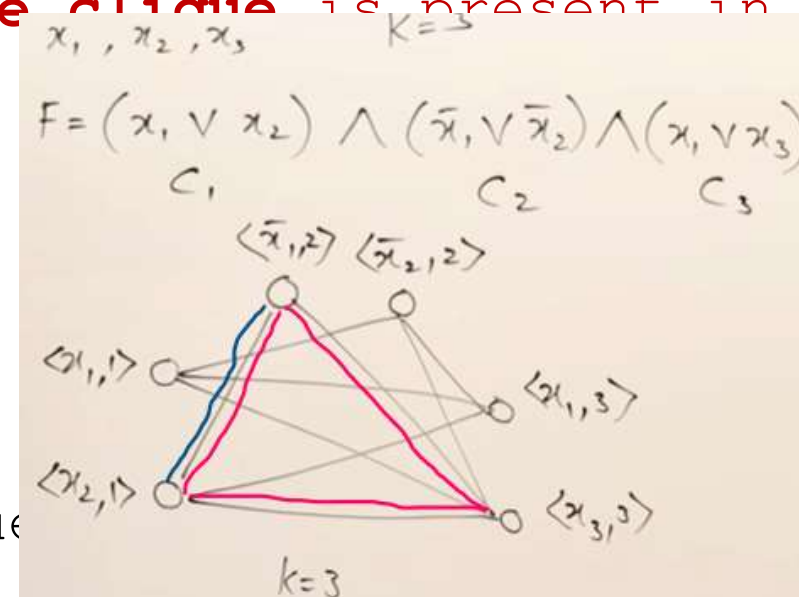
Proved here: If we take the formula **F** of **k clauses** then **k size clique** will be there in the graph.

Reverse thing also can be solved. I.e. If **K size clique** is present in graph, then Formula **F** can have **k clauses**.

- How many variable do we have? x_1, x_2, x_3
- Check the selected vertices (pink color).
- Mark them as TRUE (1)
- Substitute these values in Formula F, F is true
- Hence proved

$$F = (\overset{0}{x_1} \vee \overset{1}{x_2}) \wedge (\overset{1}{\bar{x}_1} \vee \overset{0}{\bar{x}_2}) \wedge (\overset{0}{x_1} \vee \overset{1}{x_3}) = 1$$

$C_1=1 \quad C_2=1 \quad C_3=1$



$$x_1 \quad x_2 \quad x_3$$

$$\begin{matrix} x_1 & x_2 & x_3 \\ 0 & 1 & 1 \end{matrix}$$

x_1' is true. Hence
 x_1 is made 0

- Try substituting the other values of other cliques (**Green color clique**) here

Summary: Clique Decision Problem

- **Clique Decision Problem** is NP-Hard problem only (not an NP Complete)